

Social Sciences & Humanities Open

Opening Minds, Fostering Reflection: A Pathway Analysis of Open-Mindedness and Reflective Thinking Skills in Science Education --Manuscript Draft--

Manuscript Number:	
Full Title:	Opening Minds, Fostering Reflection: A Pathway Analysis of Open-Mindedness and Reflective Thinking Skills in Science Education
Article Type:	Full Length Article
Section/Category:	Education
Keywords:	Open-mindedness; Actively Open-minded Thinking; Reflective Thinking Skill; Critical Reflection; science learning
Corresponding Author:	Pipih Nurhayati Universitas Pendidikan Indonesia INDONESIA
Corresponding Author Secondary Information:	
Corresponding Author's Institution:	Universitas Pendidikan Indonesia
Corresponding Author's Secondary Institution:	
First Author:	Pipih Nurhayati
First Author Secondary Information:	
Order of Authors:	Pipih Nurhayati Ari Widodo Widi Purwianingsih Achmad Samsudin
Order of Authors Secondary Information:	
Manuscript Region of Origin:	INDONESIA
Abstract:	Open-mindedness has been extensively researched and significantly plays a role as a moderator in students' reasoning abilities, yet its correlation or role in reflective thinking skills has not been identified. This study aims to examine the correlation and the model of reflective thinking skills influenced by open-mindedness. Open-mindedness was analysed to identify pre-dictors of students' critical reflection skills in science learning. This research employed two instruments in the form of question-nares. The first instrument measures open-mindedness using the Actively Open-Minded Thinking (AOT) disposition, consisting of three indicators: Tendency Toward Reflection, Belief Toward Revision Items, and Concern for Alternative Explanation. The second instrument, the Reflective Thinking Skill Scale for Young Science Learners, integrated the topic of photosynthesis, comprising four indicators analysed in-depth: Habitual Action, Understanding, Reflection, and Critical Reflection. Data analysis was conducted using three methods: correlation analysis, path analysis based on SEM, and influence analysis using logistic regression. The study found significant correlations observed in most indicators of reflective thinking skills. The model influencing actively open-minded thinking revealed that AOT had a powerful effect on Understanding (0.26) Reflection (0.36) and Critical Reflection (0.29). Based on the results of logistic regression beta values, it was found that tendency toward reflection significantly and most strongly (3.3) influenced reflective thinking skills. This research could serve as a reference for educators interested in fostering reflective thinking skills, particularly in science education, by considering open-mindedness as a predictor that warrants attention.
Suggested Reviewers:	Hasan Kapici

	Bogazici Universitesi hasanozgur.kapici@bogazici.edu.tr I believe that his experience in science learning is able to provide input and suggestions for the improvement of my manuscript.
	Eva Janssen Universiteit Utrecht e.m.janssen@uu.nl Her writing has inspired many of my manuscripts. Hopefully there will be an opportunity to be reviewed by her.
	Na Zhai Jinan University School of Translation Studies zhaina999@163.com
	Hafiz muhamad ihsan zafeer Shaanxi Normal University samramaqbool@zjnu.edu.cn
	Mahdi Mirhorseini Concordia University mahdi.mirhoseini@concordia.ca
Opposed Reviewers:	
Additional Information:	
Question	Response
Please confirm that you have mentioned all organizations that funded your research in the Acknowledgements section of your submission, including grant numbers where appropriate.	Yes, I confirm

Opening Minds, Fostering Reflection: A Pathway Analysis of Open-Mindedness and Reflective Thinking Skills in Science Education

Open-mindedness has been extensively researched and significantly plays a role as a moderator in students' reasoning abilities, yet its correlation or role in reflective thinking skills has not been identified. This study aims to examine the correlation and the model of reflective thinking skills influenced by open-mindedness. Open-mindedness was analysed to identify pre-dictors of students' critical reflection skills in science learning. This research employed two instruments in the form of question-naires. The first instrument measures open-mindedness using the Actively Open-Minded Thinking (AOT) disposition, consisting of three indicators: Tendency Toward Reflection, Belief Toward Revision Items, and Concern for Alternative Explanation. The second instrument, the Reflective Thinking Skill Scale for Young Science Learners, integrated the topic of photosynthesis, comprising four indicators analysed in-depth: Habitual Action, Understanding, Reflection, and Critical Reflection. Data analysis was conducted using three methods: correlation analysis, path analysis based on SEM, and influence analysis using logistic regression. The study found significant correlations observed in most indicators of reflective thinking skills. The model influencing actively open-minded thinking revealed that AOT had a powerful effect on Understanding (0.26) Reflection (0.36) and Critical Reflection (0.29). Based on the results of logistic regression beta values, it was found that tendency toward reflection significantly and most strongly (3.3) influenced reflective thinking skills. This research could serve as a reference for educators interested in fostering reflective thinking skills, particularly in science education, by considering open-mindedness as a predictor that warrants attention.

Keywords: Open-mindedness; Actively Open-minded Thinking; Reflective Thinking Skill; Critical Reflection; Science Learning

1. Introduction

How can one learn science if they do not have an open mind towards new discoveries. Learning science requires humility in learning (Marangio et al., 2023), as it involves accepting new ideas, initiating innovative actions, and is typically conducted in groups, which necessitates acceptance and interaction with new people. These three characteristics form the basic attitude of openness (Marshall et al., 2021; Souefi & Souefi, 2021). Since the 2000s, educational research has revealed that science learning can serve as a vehicle for enhancing students' thinking skills, such as critical thinking (Suparman et al., 2023), creative thinking and reflective thinking (Yan et al., 2023). This is where the bridge lies; Dewey stated that open-mindedness is a key character trait needed when one engages in reflection (Karlström & Hamza, 2023). In theory, open-mindedness and reflective thinking are closely related. However, the pathway between these two variables, especially in the context of science learning, has yet to be identified.

Learning science is akin to a miniature version of navigating life when students grow up and join society. As adults, students will encounter new situations and problems that demand solutions. The more frequently students practise reflective thinking, the more adept they become at contemplating, thinking through, and evaluating the solutions they devise (Zhai et al., 2023). Many experts assert that reflection is crucial for professional life (Adejimi et al., 2022; Shavit & Moshe, 2019; Tkavashvili, 2021). Previous research has highlighted the importance of reflective thinking as a component of an individual's metacognition. This research found that reflective journals play a role in enhancing pre-service teachers' self-awareness (Shavit & Moshe, 2019). In line with this finding, the use of diaries with a reflective approach has been found to improve communication skills, reflective thinking skills, and teachers' professional skills (Tkavashvili, 2021). However, there has been a shift in the focus of reflection research. The target is no longer solely professionals, but it has expanded to include students who need reflective thinking skills as a form of training for facing the future (Arici & Yilmaz, 2023; Leung, 2022). It can be assumed that a teacher with reflective thinking skills will contribute to developing their students' reflective skills. Ultimately, students who practise reflective thinking skills will be able to grow with the character of problem solvers and lifelong learners.

A significant amount of research has been conducted to enhance students' reflective thinking skills in science education. Several studies have found that the use of educational media, the implementation of innovative learning models, or the practice of journaling in the classroom can develop students' reflective thinking skills (Broisin et al., 2017; Yan et al., 2023; Yuen Lie Lim, 2011). One study utilised social media to enhance reflective thinking skills, using YouTube to post a series of videos. The results indicated that the use of linear videos successfully built students' reflective thinking skills and their engagement in learning (Lebedev & Sharma, 2019). Another study highlighted the importance of a structured reflection process, providing students with the opportunity to reflect on their learning process. This research was conducted on science concepts in primary schools using multimedia based on augmented reality (Chen, 2020). Researchers are beginning to view reflective thinking skills as essential not only for teachers or medical staff who require quick decision-making but also for students who are fundamentally developing these skills.

The body of educational research on reflective thinking skills continues to grow over time. Currently, several studies have conducted correlational research. Findings indicate that reflective thinking positively correlates with problem-solving. A study conducted on 36 seventh-grade students with nature science content revealed that decision-making significantly correlates positively with reflective thinking skills (Aladağ et al., 2021). Another study found that e-portfolios completed by pre-service teachers could enhance reflective thinking skills along with student motivation (Ayaz & Gök, 2022) self-regulation (Öztürk, 2020) and learning autonomy (Engeness, 2020). In these correlational studies, reflective thinking acts as the cause, with subsequent variables being the effect. For instance, an individual with strong reflective thinking skills will positively impact their academic achievement. Presently, the aim of this research is to determine what can influence students' reflective thinking skills in the absence of any intervention. One key theory utilised is Dewey's 1933 findings, which identified three main characteristics influencing the reflection process, one of which is the level of an individual's open-mindedness. Reflective thinking skills are part of the metacognitive structure, which is linked to both the cognitive and affective domains of students. Research continues to evolve to fill the knowledge gap about the role of reflective thinking in supporting students' learning processes.

Previous research has indicated that open-mindedness is considered an inquiry virtue (Marangio et al., 2023). Open-mindedness allows students to consider potential possibilities, interpret past events, and re-evaluate their personal assumptions. A study involving 47 participants found that individuals who are open-minded can accept new ideas, perspectives, and questions. Conversely, close-minded individuals struggle to accept suggestions and achieve consensus within a group (Karlström & Hamza, 2023). During reflection and discussion, open-mindedness plays a crucial role in making discussions either constructive and collaborative or competitive and disputatious. Future research is needed to analyse individual differences, such as humility in learning, as open-mindedness can significantly influence a student's reasoning process.

Open-mindedness

In line with the development of educational research, open-mindedness is increasingly recognised as a fundamental component in learning, encouraging students to explore new ideas and accept new perspectives. Open-minded thinking is often described as forming one's opinions based on evidence and reasoned arguments. However, in reality, we all hold numerous beliefs that necessitate relying on the expertise of others, including scientists who routinely engage in this practice (Harding & Hare, 2000). Open-mindedness requires the capacity to consider and reflect on different viewpoints, seek evidence that may contradict existing beliefs, and accept new ideas and possibilities. Open-minded thinking is acknowledged as one of the dispositions that can influence how an individual rationalises arguments and makes decisions. Active open-minded thinking is one of various thinking dispositions (Stanovich & Toplak, 2023). These dispositions, along with differences in individual cognitive abilities, play a crucial role in explaining the extent to which someone is inclined towards rational thinking (Janssen et al., 2020a). Common understandings of

open-mindedness and its three key characteristics include: Willingness to relinquish views, Open-mindedness involves readiness to abandon one's beliefs when presented with strong evidence that they are untrue [22], [23], [25]; Fundamental openness to new things, Individuals who are open-minded are naturally receptive to new ideas, innovative actions, and interactions with new people [24], [26]; and, Intellectual virtue, Open-mindedness is seen as an intellectual virtue, meaning it is a positive character trait that promotes the acquisition of epistemic goods, which include knowledge and understanding. These core attributes collectively define the concept of open-mindedness and its significance in the realm of thought and intellectual growth across various disciplines, including science education.

Reflective Thinking

In contemporary society, individuals are expected to possess a variety of skills, including problem-solving, creative thinking, critical thinking, learning to learn, reflective thinking, and the development of metacognitive variables (Kirbulut & Uzuntiryaki-Kondakci, 2019; Orakcı, 2021; Sebatana & Dudu, 2022). Additionally, due to the rapid advancements in educational technology in the 21st century and the transformative period, learners are required to acquire skills to become independent reflective thinkers as part of this paradigm shift (Cirak Kurt & Yildirim, 2021; Muhammad et al., 2023; van Velzen, 2015). Among all these skills, reflective thinking is considered the most significant because it is a higher-order thinking skill that involves recognising problems, reflecting on actions taken to solve problems, and analysing what was done well or poorly for further improvement (Menon & Azam, 2021; Onrubia et al., 2022; Orcid et al., 2023). This particular skill demands that individuals consistently re-evaluate and adapt their problem-solving strategies to discover new ways of gathering information (Ayaz & Gök, 2022; Bunt & Gouws, 2020; Maksimović & Osmanović, 2019).

Reflective thinking skills form the foundation of self-directed learning (Slepcevic-Zach & Stock, 2018). These skills enable students to recall, reflect on, and critically evaluate their learning processes (David Kember et al., 2000). They encourage the development of critical reflection skills, which involve recognising, understanding, and accepting paradigm shifts as knowledge grows (Hoggan & Kloubert, 2020). Previous research has categorised the structure of reflective thinking skills from non-reflectors to critical reflectors. Non-reflectors are characterised by knowledge based on intuition and observation without linking it to concepts, thus facing challenges in formulating innovations based on their understanding. Conversely, reflectors can generate innovations and paradigm shifts that influence their behaviour. To map these achievements, indicators of reflective thinking skills, including Habitual Action, Understanding, Reflection, and Critical Reflection, are used (David Kember et al., 2000). Reflective thinking skills support learning outcomes, performance, attitudes, and provide added value in the workplace (Cirak Kurt & Yildirim, 2021; Hutner et al., 2021). For example, a non-reflective teacher may not modify their lesson plans despite unsatisfactory student learning outcomes, ultimately affecting the quality of student learning and teaching. Similarly, in the medical field (Wallace et al., 2013), doctors need to reflect to make accurate diagnoses for their patients. However, the role and influence of active open-minded thinking have not been fully explored in relation to reflective thinking skills, particularly in science education.

Open-mindedness and Reflective Thinking in Science Learning

Scientific learning evolves based on new theories or principles, and other findings suggest that open-mindedness correlates with attitudes towards knowledge (Svedholm-Häkkinen & Lindeman, 2018). This is supported by the fact that an open-minded character is essential in science education (Harding & Hare, 2000). Without open-mindedness, the abilities and knowledge of scientists can stagnate and fail to develop. When new ideas are accepted, this facilitates the reflection process, leading to new perspectives, behaviours, and readiness to apply knowledge as a result of reflection (van Velzen, 2015). For instance, one scientific topic that demands open-mindedness and reflection

is photosynthesis. Through conceptual analysis, photosynthesis stands as a core concept linking metabolism, the carbon cycle, the hydrogen cycle, and forming the basis of the food chain. Open-mindedness is crucial given the discoveries about the mechanisms and factors affecting photosynthesis. Meanwhile, reflective thinking is essential for students to delve deeper into the fundamental aspects of the photosynthesis process for the sustainability of life.

Open-mindedness is also crucial when students manage information and trigger reflection (Baron, 2017). The classical reflection cycle involves stages of experience, reflection, and outcome. At the experience stage, students acquire information, behaviours, or feelings. This experience is then reflected upon in various ways, such as reconsidering the encounter and re-evaluating it based on their experiences. The outcomes derived from this reflective process include new knowledge, behaviours, and even commitments to take concrete actions (Muhammad et al., 2023). These activities require specific thinking skills known as reflective thinking skills.

However, individuals vary in their level of open-mindedness. It is impossible to force someone to adopt an open character. This trait becomes a cognitive style that individuals use to develop their knowledge (Marshall et al., 2021). This aligns with findings that there is also individual variation in the motivation to engage in rational thinking processes (Stanovich, 2018). Thus, the literature review found focuses on how open-mindedness affects various student abilities and skills. Open-mindedness is closely related to rational judgment, including the ability to avoid knowledge bias and confirmation bias (Marshall et al., 2021). Research has shown that open-mindedness is linked to how individuals argue (Svedholm-Häkkinen & Lindeman, 2018) and make decisions (Mirhoseini et al., 2023a). However, there is a lack of empirical knowledge on how open-mindedness affects students' reflective thinking skills in science learning. This is important to research as it can provide educators with insights into improving skills based on individual differences that often occur (Wong et al., 1995). Figure 1 illustrates the hypothesised relationship between Actively Open-minded Thinking (AOT) and Reflective Thinking Skills (RTS). The RTS indicators studied include Habitual Action, Understanding, Reflection, and Critical Reflection (D Kember et al., 2000).

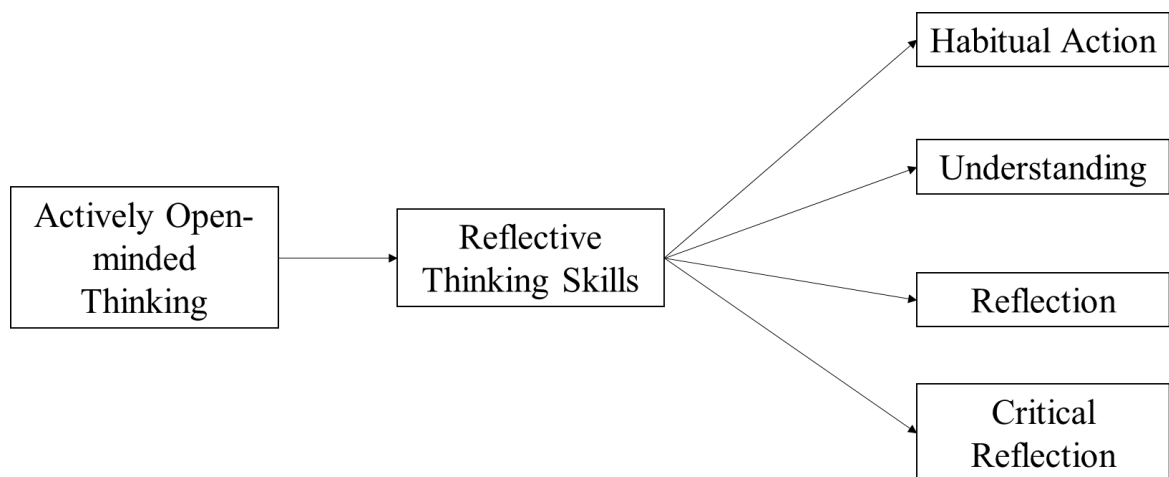


Fig 1. Hypothesized Pathway of Actively Open-minded Thinking and Reflective Thinking Skills

Based on the background, this research aims to explore the role and influence of open-mindedness on students' reflective thinking skills empirically. Therefore, this study will first examine the correlation and model of open-mindedness concerning the elements of reflective thinking skills. Secondly, it will investigate the role of the elements of open-mindedness as predictors in students' reflective thinking skills. These two objectives are articulated in the research questions as follows:

1. How does the correlation and model of actively open-minded thinking relate to the elements of re-lective thinking skills (habitual action, understanding, reflection, and critical reflection)?

2. How do the elements of open-mindedness function as predictors in students' reflective thinking skills?

2. Method

Instruments used

This study employed two primary instruments. To gauge the level of open-mindedness, the Actively Open-minded Thinking (Janssen et al., 2020) scale was utilized. It comprises 30 items categorized into three main indicators: tendency toward reflection, concern for alternative explanations, and belief on revision items. The validity and reliability of this instrument have been examined in various studies (Baron, 2017; Stanovich & Toplak, 2023).

Table 1. Actively Open-Minded Thinking Indicators and Sample Statements

Indicators	Statements
Tendency toward reflection	Changing your mind is a sign of weakness.
	It is important to stick to our opinion even if there is evidence against it.
	Considering too many different opinions will confuse our minds.
Concern For Alternative Explanations	It is better to listen to other opinions than to stick to your own opinion.
	We do not need to think about new things, just with what we know now.
	Confused thoughts can occur if we think too much about different things.
Belief Revision Item	Even though there are friends who disagree, we must stick to our opinion.
	When there is evidence that contradicts our opinion, we need to stay strong in our opinion.
	It is important to consider new information that may come.

Secondly, an instrument comprising 36 questions was used to measure reflective thinking skills. These questions embody hierarchical elements adapted from the Reflective Thinking Skill Scale (Kember et al., 1999) and were developed into the Reflective Thinking Skill for Science Young Learner (RTSYL) Scale. This instrument was conceptualized based on the photosynthesis science concept and translated into Indonesian.

Table 2. Reflective Thinking Skills Indicators and Sample Statements

Indicators	Statements
Habitual Action	When the teacher starts explaining, I immediately take notes in my notebook.
	I underline sentences or words that I consider important in the book.
	I can recall the material through the teacher's explanations.
Understanding	I can explain how temperature plays a role in the process of photosynthesis.
	I have a good understanding of how plants use sunlight in photosynthesis.
	I know that photosynthesis is the process that converts sunlight into food for plants.
Reflection	I critically evaluate the sources of information I use to understand photosynthesis.
	I try to compare my previous knowledge with the new information I learn.
	I actively seek additional resources and the latest information to deepen my understanding.
Critical Reflection	I did not know that plants take carbon dioxide from the air to make food.
	Initially, I did not understand how sunlight helps plants grow, but now I know that plants need sunlight.
	Until now, I am not sure why photosynthesis plays an important role in addressing climate change.

The elements within the RTSYL encompass habitual action (HA) during science learning, understanding of scientific concepts, reflection during the learning process, and critical re-flection when applying acquired concepts. The internal instrument reliability was tested using Cronbach Alpha (AOT: 0.870, RTSYL: 0.947).

Participants in the study

Ninety students aged 12-13 participated in this study, primarily from the 7th grade or equivalent level in junior high school. The participants consisted of 41 male students and 49 female students. Most students hailed from diverse areas, with 46% residing in border areas, 49% in city centers, and the remainder from rural regions. These students were enrolled in two types of schools: public schools and religious-based schools. Despite the distinct nature of these school bases, both institutions maintained equivalent science curriculum and learning outcomes.

Statistical Analysis

In this research, three distinct statistical analyses were utilised with differing objectives. Initially, multivariate analysis of variance alongside univariate tests was employed to explore the correlation between actively open-minded thinking and elements of reflective thinking skills. Secondly, to examine the model of how open-mindedness influences reflective thinking skills, a path analysis was conducted using structural equation modelling (SEM) facilitated by the SPSS-AMOS 23 application. Path analysis categorizes effects from small influence to powerful impact (Keith & Cool, 1992), as follows in the Table 3.

Table 3. Path value analysis categorization

Value of path	Categorization
0.05 - 0.10	Small influence (but meaningful effects)
– 0.25	Medium influence
Up to 0.25	Large influence (powerfull effects)

To identify a model that best represents the influence of open-mindedness, various indicators including goodness of fit index (GFI), comparative fit index (CFI), Tucker Lewis index (TLI), root mean square error of approximation (RMSEA), and standardized root mean square residual (SRMR) were analysed to assess model fit. Lastly, logistic regression analysis was employed to determine the factors within reflective thinking skills most impacted by open-mindedness.

3. Results

Correlation test

The initial analysis involved testing the correlation between the variable of actively open-minded thinking and the indicators of reflective thinking skills, namely habitual action during learning, understanding of the material, reflection, and critical reflection. The correlations among these elements varied, indicating both weak and strong associations. Table 4 demonstrates that the correlations among the elements of reflective thinking skills ranged significantly between 0.259 and 0.603. Notably, there was an insignificant correlation between open-mindedness and habitual action (0.19). The highest correlation observed was between habitual action and reflection (0.721).

Table 4. Correlation among Actively Open-Minded Thinking and Reflective Thinking Skills

Elements	1	2	3	4	5
----------	---	---	---	---	---

Actively Open-minded Thinking	-	0.019	0.208*	0.562*	0.253*
Habitual Action Understanding	-	0.560**	0.721**	0.562*	
Reflection	-	0.591*	0.259**		
Critical Reflection	-	0.500**			
**p < .01, * p < .05					

The data above shows that Actively Open-minded Thinking (AOT) does not have a significant correlation ($r = 0.0019$). However, AOT has a significant positive correlation with Understanding (0.208^*), Reflection ($r = 0.562^*$), and Critical Reflection ($r = 0.253^*$). Based on the analysis of the data, it can be assumed that students who exhibit a tendency towards open-minded thinking are able to have a better understanding and critically reflect on their actions. Understanding has a significant correlation with Reflection ($r = 0.591^*$) and Critical Reflection ($r = 0.259^{**}$). Reflection has a significant correlation with Critical Reflection ($r = 0.500^{**}$).

Test of the model

To test the proposed model hypothesis, an analysis based on SEM path analysis was conducted. Each pathway in the model signifies an "influence" value. The categories are divided into small influence with a path value of 0.05-0.10, which, although small, holds meaningful effects; moderate influence with a value of 0.11-0.25; and large influence with a value above 0.25, indicating powerful effects (Keith & Cool, 1992). According to these references, it was found that open-mindedness holds a powerful effect on students' understanding (0.26) of concepts and their critical reflection (0.29). Similarly, within the elements of reflective thinking skills, habitual action showed a powerful effect on understanding (0.36) and critical reflection (0.26). Moreover, understanding displayed a powerful effect on reflection (0.30) and critical reflection (0.40). Moderate influence was observed in the relationship between habitual action and reflection (0.10) as well as critical reflection (0.19). Although small, open-mindedness had a significant influence on habitual action (0.4).

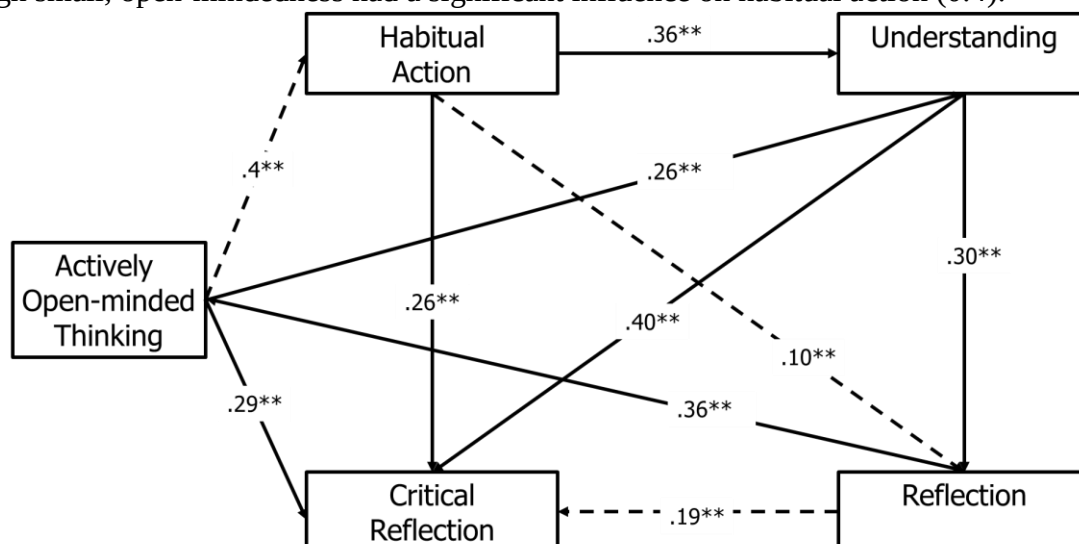


Figure 2. Indonesian reflective thinking skill influence by actively open-minded thinking model
RMSEA: 0.624 CFI: 0.60 GFI: 0.758 TLI: 0.600

Examining the effect of actively open-minded thinking factor on reflective thinking skill

The Logistic Regression analysis was conducted to understand the influence of each element of actively open-minded thinking on critical reflection skills. The dependent variable in this analysis was the critical reflection skills concerning the topic of photosynthesis in science, which were tested and included the ability to recognize changes in personal perspectives, modifications in how things are done based on their understanding of the concept of Photosynthesis, and identifying shortcomings in previously held beliefs about Photosynthesis.

Table 5. Actively open-minded thinking influencing on students' critical reflection skill

Elements	B	S. E	Wald	Sig.	Odds ratio
Tendency Toward Reflection	1.19	0.44	7.25	.00	3.3
Concern For Alternative Explanation	-1.41	0.52	7.34	.00	0.24
Belief of Revision Item	0.42	0.41	1.04	.30	1.53

Nagelkerke R²=0.19

Table 5 displays the beta coefficients from logistic regression on each indicator of actively open-minded thinking influencing critical reflection skills. Tendency Toward Reflection (B= 1.19, p <0.001) exhibits a positive association with critical reflection skills, whereas Concern for Alternative Explanation (B= -1.41, p <0.001) shows a negative correlation with critical reflection skills.

The tendency to reflect has a significant positive influence on Critical Reflection (B= 1.19). Wald 7.25 indicates that this result is significant. With a p-value less than 0.05, it shows that the influence is highly significant. The odds ratio of 3.3 indicates that students who tend to reflect are 3.3 times more likely to have Critical Reflection skills. Concern for alternative explanations has a negative influence on the ability of critical reflection (B= -1.14). Wald 7.34 indicates that this result is significant. The odds ratio is quite large, indicating that students who are concerned have a 24% lower chance of developing their Critical Reflection skills. Belief in revision items has a positive but nonsignificant influence based on the Wald and Sig. values being equal to 0.30. The odds ratio indicates that students who believe in revision items are 1.53 times more likely to have Critical Reflection skills, but the result is not significant. The Nagelkerke R² indicates the logistic regression in assessing how well the model explains the variability of the given data. The value of 0.19 indicates a moderate level for social research, which tends to be influenced by many factors.

4. Discussions

Actively Open-minded Thinking Correlation and The Pathway

Path analysis results found a weak but significant influence on students' habitual actions and reflections. Habitual action, as spontaneous behavior such as taking notes when someone explains or underlines notes when encountering important concepts, is not significantly influenced by their level of open-mindedness. Previous research has shown that habitual actions have the smallest correlation because this skill is more mechanical and does not involve reflective processes (Ghanizadeh, 2017). Habitual actions are non-reflective activities that can be recognised through student answers that show no attempt to understand the underlying concepts or theories related to the topic.

Actively Open-minded Thinking has a positive correlation with Understanding, Reflection, and Critical Reflection. This finding shows that open-mindedness can motivate someone to reflect. At the same time, there is a strong relationship between the indicators of reflective thinking, namely Understanding, Reflection, and Critical Reflection. Understanding indicator involves understanding concepts and the ability to apply them in different contexts (Zach & Ophir, 2020). As a characteristic of science learning that continues to evolve due to new discoveries, actively open-minded thinking significantly affects the understanding of scientific topics. Understanding refers to students showing evidence of understanding the concept or topic. This is in line with the need for acceptance of new

ideas and theories in science learning (Marangio et al., 2023). It is also consistent with the theory that cognitive processes such as reflective thinking are supported by the values one believes in (Bialystok & Ferkany, 2019; Curzer & Gottlieb, 2019). Therefore, the presence of AOT mutually influences the understanding of dynamic and evolving scientific concepts if they believe in them.

The next results indicate that AOT has a fairly high correlation with Reflection. Students reflecting on what they have done by recalling, pondering, and evaluating their learning processes, significantly influenced by students' actively open-minded thinking. Reflection tends to guide students to recall and evaluate the learning process to formulate action plans (Tanyeli Zeki & Kuter, 2018). In the researched reflection process, the reflection carried out considers information, new facts, and other people's opinions. Additionally, students are asked about how they evaluate their learning independently. Reflection is demonstrated when students have the willingness and habit to consider and reflect on better ways of doing things. Reflection is a situation where students contemplate, think, and evaluate their understanding of a concept, situation, or condition. These findings are important to follow up, especially among adolescents in Indonesia, where in enhancing reflective thinking skills, students need support in the reflection process. Some experts express that the reflection stage requires curiosity and a responsive attitude (Marshall et al., 2021). As input for further research, individual measurement at the level of students' curiosity and responsiveness is needed. Thus, it can complement the knowledge gap about the construct of open-mindedness in accommodating and facilitating the reflective process optimally.

Similarly, the path analysis results show that critical reflection is significantly influenced by AOT. Critical Reflection involves the ability to respond to paradigm shifts and demonstrate behavior that aligns with one's understanding (David Kember et al., 1999). The ultimate goal of critical reflection is for learners to understand and interpret (Orcid et al., 2023). This is consistent with previous research indicating that actively open-minded thinking influences a person's attitudes and knowledge (Mirhoseini et al., 2023a; Stanovich & Toplak, 2019). This model shows that open-mindedness plays a crucial role in facilitating students' mastery of reflective thinking skills, especially their level of understanding reflection and ability for critical reflection.

This disposition is closely related to rationality. Some psychological studies have found that the tendency to think rationally is influenced by engagement and willingness to think; this disposition can be measured using the Need for Cognition scale. Other dispositions that affect rational thinking include the Rational Experiential Inventory scale and the scale of considering future consequences (Jansen & Spitzer, 2009; Toplak et al., 2014). Among the various thinking dispositions, actively open-minded thinking plays a significant role in the development of rational thinking (Janssen et al., 2020a). Actively open-minded thinking can indicate intelligence and the level of rationality of an individual (Baron et al., 2015). Someone who has the characteristic of always seeking various perspectives tends to show more rational behavior (Janssen et al., 2020b; Svedholm-Häkkinen, 2009). Others' opinions will be considered before forming their own views (Mirhoseini et al., 2023b). It can be assumed that intelligence and rational thinking develop because of the willingness to accept various perspectives and then form their own views based on feedback and other information they receive.

Someone with high AOT tends to seek evidence before accepting new information, tends to continue seeking information from various perspectives, and is not easily discouraged (Mirhoseini et al., 2023b). Additionally, the high tendency of AOT in someone is influenced by tolerant characteristics and the ability to accept new facts (Sedl, 2023). Conversely, someone closer to a close-minded character is not able to accept new views and tends to reject and only believe in their own views (Mirhoseini et al., 2023b). An interesting finding is that open-mindedness is the key to reflection in an individual (Svedholm-Häkkinen & Lindeman, 2018). Reflection that occurs during a discussion built on openness will result in constructive discussion rather than competition. In science learning, students are confronted with new concepts or facts that need to be examined more deeply.

The role of open-mindedness factor on reflective thinking skill

In line with the theory that reflective thinking is a state of mind and an active engagement process with an object of investigation, reflection (Marshall et al., 2021), according to Leboskey, has three main factors: open-mindedness, wholeheartedness, and responsibility (Karlström & Hamza, 2023). The extent to which open-mindedness is empirically studied as a mediator in reflection is not yet well-researched. There are four characteristic features of open-mindedness: the ability to accept new ideas, consider current and past issues, examine and seek alternative solutions, and have a sense of curiosity (Marshall et al., 2021). This is highly relevant to science learning, which requires an open attitude and the ability to accept new facts. Reflective practices that have been extensively studied, such as journal or learning diary writing, require honesty and openness (Adejimi et al., 2022; Muhammad et al., 2023). Consistent with this, reflective thinking requires wholeheartedness and is grounded in curiosity.

The findings from logistic regression can be interpreted to indicate that students with a low tendency toward reflection may find it more challenging to apply critical reflection during learning. The tendency toward reflection emerges as an important predictor in achieving students' reflective thinking skills. Therefore, it can be said that students who tend to reflect are likely to master critical reflection skills. This finding is similar to research indicating that a tendency toward reflection can counter fake news through reflection (Mirhoseini et al., 2023a). On the other hand, other variables did not emerge as predictors of reflective thinking skills. Concern for Alternative Explanations and Belief in Revision Item did not affect how students develop their critical reflection skills.

Previous research has emphasized the important influence of personal factors such as learning, and open-mindedness plays a crucial role in enhancing learning. This statement is reinforced because open-mindedness can promote knowledge acquisition, adaptation, and information sharing among human resources (Al-Abrow et al., 2023). It also leads to the side effect of humility to continue learning and a willingness to listen to others' advice (Bialystok & Ferkany, 2019).

Limitation and future research

The characteristic of open-mindedness may be the key for educators who want to enhance students' reflective thinking abilities in science education. This is a fundamental aspect for educators who aim to increase students' willingness to accept new facts in science learning and to engage in reflection as part of the learning process. This phenomenon requires further investigation, especially to understand the role of open-mindedness in the level of students' reflective thinking abilities in science education.

This research has limitations in the data collection process, as the data were collected simultaneously, potentially causing cause-and-effect effects on the obtained data. Open-mindedness is a virtue that may originate from cultural influences or be influenced by family treatment, environmental factors, and school. The reflective thinking instrument used in this study, focused on science learning specifically on the topic of photosynthesis, might yield better results if tested comprehensively. The limitations of this research could be an opportunity for other fields of study and could fill existing gaps. The analysis primarily focused on the end result, possibly overlooking supporting factors as mentioned above, which were not clearly captured in this study. Subsequent research could consider adding more variables such as students' curiosity, motivation, learning environment, social support, and others. This would provide an overview and explain the variability that influences the model developed.

Conclusions

This study aims to uncover the influence and role of open-mindedness, measured through actively open-minded thinking disposition, on the structure of reflective thinking skills in science education.

The impact and role of actively open-minded thinking are revealed through correlation tests and model assessments using SEM-based path analysis. As anticipated, actively open-minded thinking shows a substantial impact on elements such as understanding and critical reflection. The findings also indicate a high correlation among each component of reflective thinking, creating a cascading effect that requires development from pre-reflective stages like habitual actions and understanding to reflective stages like reflection and critical reflection. This provides valuable insights for educators to further develop reflective thinking skills during science learning activities in the classroom.

Another finding from the logistic regression analysis highlights that the most influential element in determining students' achievement in critical reflection skills is the tendency to reflect. The ability for critical reflection tends to increase in line with students' tendency to reflect during the learning process. Based on these findings, it is recommended to embed reflective activities in the learning environment, encourage an increase in students' tendency to reflect, and cultivate a culture of good critical reflective habits.

References

- Adejimi, S. A., Nzabalirwa, W., & Shivoga, W. A. (2022). Enhancing Students' Attitudes Toward Biology Using Consensus and Cooperative Reflective Journal Writing Educational Strategies. *Problems of Education in the 21st Century*, 80(2), 242–255. <https://doi.org/10.33225/pec/22.80.242>
- Al-Abrow, H., Fayez, A. S., Abdullah, H., Khaw, K. W., Alnoor, A., & Rexhepi, G. (2023). Effect of open-mindedness and humble behavior on innovation: mediator role of learning. *International Journal of Emerging Markets*, 18(9), 3065–3084. <https://doi.org/10.1108/IJOEM-08-2020-0888>
- Aladağ, E., Arıkan, A., & Özenoğlu, H. (2021). Nature education: Outdoor learning of map literacy skills and reflective thinking skill towards problem-solving. *Thinking Skills and Creativity*, 40. <https://doi.org/10.1016/j.tsc.2021.100815>
- Arici, F., & Yilmaz, M. (2023). An examination of the effectiveness of problem-based learning method supported by augmented reality in science education. *Journal of Computer Assisted Learning*, 39(2), 446–476. <https://doi.org/10.1111/jcal.12752>
- Ayaz, M., & Gök, B. (2022). The effect of e-portfolio application on reflective thinking and learning motivation of primary school teacher candidates. *Current Psychology*.
- Baron, J. (2017). Comment on Kahan and corbin: Can polarization increase with actively open-minded thinking? *Research and Politics*, 4(1). <https://doi.org/10.1177/2053168016688122>
- Baron, J., Scott, S., Fincher, K., & Emlen Metz, S. (2015). Why does the Cognitive Reflection Test (sometimes) predict utilitarian moral judgment (and other things)? *Journal of Applied Research in Memory and Cognition*, 4(3), 265–284. <https://doi.org/10.1016/j.jarmac.2014.09.003>
- Bialystok, L., & Ferkany, M. A. (2019). Open-Mindedness from the Public Sphere to the Classroom. *Educational Theory*, 69(4), 377–381. <https://doi.org/10.1111/edth.12375>
- Broisin, J., Venant, R., & Vidal, P. (2017). Awareness and reflection in virtual and remote laboratories: The case of computer education. *International Journal of Technology Enhanced Learning*, 9(2–3), 254–274. <https://doi.org/10.1504/IJTEL.2017.084503>
- Bunt, B., & Gouws, G. (2020). Using an artificial life simulation to enhance reflective critical thinking among student teachers. *Smart Learning Environments*, 6.
- Chen, C. H. (2020). Impacts of augmented reality and a digital game on students' science learning with reflection prompts in multimedia learning. *Educational Technology Research and Development*, 68(6), 3057–3076. <https://doi.org/10.1007/s11423-020-09834-w>
- Cirak Kurt, S., & Yildirim, I. (2021). The effects of blogging on pre-service teachers' reflective thinking and self-efficacy. *Reflective Practice*, 22(2), 233–249. <https://doi.org/10.1080/14623943.2021.1879772>
- Curzer, H. J., & Gottlieb, J. (2019). Making the Classroom Safe for Open-Mindedness. *Educational Theory*, 69(4), 383–402. <https://doi.org/10.1111/edth.12376>

- Engeness, I. (2020). Teacher facilitating of group learning in science with digital technology and insights into students' agency in learning to learn. *Research in Science and Technological Education*, 38(1), 42–62. <https://doi.org/10.1080/02635143.2019.1576604>
- Ghanizadeh, A. (2017). The interplay between reflective thinking, critical thinking, self-monitoring, and academic achievement in higher education. *Higher Education*, 74(1), 101–114. <https://doi.org/10.1007/s10734-016-0031-y>
- Harding, P., & Hare, W. (2000). Portraying science accurately in classrooms: Emphasizing open-mindedness rather than relativism. *Journal of Research in Science Teaching*, 37(3), 225–236. [https://doi.org/10.1002/\(SICI\)1098-2736\(200003\)37:3<225::AID-TEA1>3.0.CO;2-G](https://doi.org/10.1002/(SICI)1098-2736(200003)37:3<225::AID-TEA1>3.0.CO;2-G)
- Hoggan, C., & Kloubert, T. (2020). Transformative Learning in Theory and Practice. *Adult Education Quarterly*, 70(3), 295–307. <https://doi.org/10.1177/0741713620918510>
- Hutner, T. L., Petrosino, A. J., & Salinas, C. (2021). Do Preservice Science Teachers Develop Goals Reflective of Science Teacher Education? A Case Study of Three Preservice Science Teachers. *Research in Science Education*, 51(3), 761–789. <https://doi.org/10.1007/s11165-018-9816-6>
- Jansen, A., & Spitzer, S. M. (2009). Prospective middle school mathematics teachers' reflective thinking skills: Descriptions of their students' thinking and interpretations of their teaching. *Journal of Mathematics Teacher Education*, 12(2), 133–151. <https://doi.org/10.1007/s10857-009-9099-y>
- Janssen, E. M., Verkoeijen, P. P. J. L., Heijltjes, A. E. G., Mainhard, T., van Peppen, L. M., & van Gog, T. (2020a). Psychometric properties of the Actively Open-minded Thinking scale. *Thinking Skills and Creativity*, 36(March), 100659. <https://doi.org/10.1016/j.tsc.2020.100659>
- Janssen, E. M., Verkoeijen, P. P. J. L., Heijltjes, A. E. G., Mainhard, T., van Peppen, L. M., & van Gog, T. (2020b). Psychometric properties of the Actively Open-minded Thinking scale. *Thinking Skills and Creativity*, 36(April), 100659. <https://doi.org/10.1016/j.tsc.2020.100659>
- Karlström, M., & Hamza, K. (2023). Modeling preservice middle school science teachers' reflective practice. *Cogent Education*, 10(1). <https://doi.org/10.1080/2331186X.2023.2215615>
- Keith, T. Z., & Cool, V. A. (1992). Testing models of school learning: Effects of quality of instruction, motivation, academic coursework, and homework on academic achievement. *School Psychology Quarterly*, 7(3), 207–226. <https://doi.org/10.1037/h0088260>
- Kember, D., Leung, D. Y. P., Jones, A., Loke, A. Y., McKay, J., Sinclair, K., Tse, H., Webb, C., Wong, F. K. Y., Wong, M., & Yeung, E. (2000). Development of a questionnaire to measure the level of reflective thinking. *Assessment and Evaluation in Higher Education*, 25(4), 381–395. <https://doi.org/10.1080/713611442>
- Kember, David, Jones, A., Loke, A., McKay, J., Sinclair, K., Tse, H., Webb, C., Wong, F., Wong, M., & Yeung, E. (1999). Determining the level of reflective thinking from students' written journals using a coding scheme based on the work of Mezirow. *International Journal of Lifelong Education*, 18(1), 18–30. <https://doi.org/10.1080/026013799293928>
- Kember, David, Leung, D. Y. P., Jones, A., Loke, A. Y., McKay, J., Sinclair, K., Tse, H., Webb, C., Wong, F. K. Y., Wong, M., & Yeung, E. (2000). Development of a questionnaire to measure the level of reflective thinking. *Assessment and Evaluation in Higher Education*, 25(4), 381–395. <https://doi.org/10.1080/713611442>
- Kirbulut, Z. D., & Uzuntiryaki-Kondakci, E. (2019). Examining the mediating effect of science self-efficacy on the relationship between metavariabls and science achievement. *INTERNATIONAL JOURNAL OF SCIENCE EDUCATION*, 41(8), 995–1014. <https://doi.org/10.1080/09500693.2019.1585594>
- Lebedev, P., & Sharma, M. D. (2019). Riddles on youtube: Investigating the potential to engage viewers in reflective thinking. *Research in Learning Technology*, 27(1063519), 1–12. <https://doi.org/10.25304/rlt.v27.2280>
- Leung, J. S. C. (2022). Shifting the Teaching Beliefs of Preservice Science Teachers About Socioscientific Issues in a Teacher Education Course. *International Journal of Science and Mathematics Education*, 20(4), 659–682. <https://doi.org/10.1007/s10763-021-10177-y>

- Maksimović, J., & Osmanović, J. (2019). Perspective of cognitive thinking and reflective teaching practice. *International Journal of Cognitive Research in Science, Engineering and Education*, 7(2), 1–10. <https://doi.org/10.5937/IJCRSEE1902001M>
- Marangio, K., Carpendale, J., Cooper, R., & Mansfield, J. (2023). Supporting the Development of Science Pre-service Teachers' Creativity and Critical Thinking in Secondary Science Initial Teacher Education. *Research in Science Education*, 0123456789. <https://doi.org/10.1007/s11165-023-10104-x>
- Marshall, T., Keville, S., Cain, A., & Adler, J. R. (2021). On being open-minded, wholehearted, and responsible: a review and synthesis exploring factors enabling practitioner development in reflective practice. *Reflective Practice*, 22(6), 860–876. <https://doi.org/10.1080/14623943.2021.1976131>
- Menon, D., & Azam, S. (2021). Investigating Preservice Teachers' Science Teaching Self-Efficacy: an Analysis of Reflective Practices. *International Journal of Science and Mathematics Education*, 19(8), 1587–1607. <https://doi.org/10.1007/s10763-020-10131-4>
- Mirhoseini, M., Early, S., El Shamy, N., & Hassanein, K. (2023a). Actively open-minded thinking is key to combating fake news: A multimethod study. *Information and Management*, 60(3), 103761. <https://doi.org/10.1016/j.im.2023.103761>
- Mirhoseini, M., Early, S., El Shamy, N., & Hassanein, K. (2023b). Actively open-minded thinking is key to combating fake news: A multimethod study. *Information and Management*, 60(3), 103761. <https://doi.org/10.1016/j.im.2023.103761>
- Muhammad, H., Zafeer, I., Li, Y., & Maqbool, S. (2023). An Approach to Progress Learning Outcomes : International Graduate Students ' Engagement in Reflective Practice and Reflective Journal Writing during Pandemic. *Sustainability*.
- Onrubia, J., Roca, B., & Minguella, M. (2022). Assisting teacher collaborative discourse in professional development: An analysis of a facilitator's discourse strategies. *Teaching and Teacher Education*, 113. <https://doi.org/10.1016/j.tate.2022.103667>
- Orakcı, Ş. (2021). Teachers' Reflection and Level of Reflective Thinking on the Different Dimensions of their teaching practice. *International Journal of Modern Education Studies*, 5(1), 117. <https://doi.org/10.51383/ijonmes.2021.88>
- Orcid, F. G. K., Berk, A., Orcid, U., Orcid, K. Z., Orcid, R. Y., & Received, T. (2023). Metacognitive Awareness , Reflective Thinking , Problem Solving , And Community Of Inquiry As Predictors Of Academic Self-Efficacy In Blended Learning : A Correlational Study. *Turkish Online Journal of Distance Education-TOJDE*, January, 20–36.
- Öztürk, M. (2020). The Relationship between Self-Regulation and Proportional Reasoning: The Mediating Role of Reflective Thinking towards Problem Solving. *Eğitim ve Bilim*, 45(204), 143–155. <https://doi.org/10.15390/EB.2020.8480>
- Sebatana, M. J., & Dudu, W. T. (2022). Reality or Mirage: Enhancing 21st-Century Skills Through Problem-Based Learning While Teaching Particulate Nature of Matter. *International Journal of Science and Mathematics Education*, 20(5), 963–980. <https://doi.org/10.1007/s10763-021-10206-w>
- Sedl, M. (2023). *Thinking dispositions are related to more positive attitudes toward immigrants*. 205(August 2022). <https://doi.org/10.1016/j.paid.2023.112100>
- Shavit, P., & Moshe, A. (2019). The contribution of reflective thinking to the professional development of pre-service teachers. *Reflective Practice*, 20(4), 548–561. <https://doi.org/10.1080/14623943.2019.1642190>
- Slepcevic-Zach, P., & Stock, M. (2018). ePortfolio as a tool for reflection and self-reflection. *Reflective Practice*, 19(3), 291–307. <https://doi.org/10.1080/14623943.2018.1437399>
- Souefi, N. El, & Souefi, N. El. (2021). in *International Schools in Egypt 2021*.
- Stanovich, K. E. (2018). How to think rationally about world problems. *Journal of Intelligence*, 6(2), 1–12. <https://doi.org/10.3390/jintelligence6020025>
- Stanovich, K. E., & Toplak, M. E. (2019). The need for intellectual diversity in psychological science: Our own studies of actively open-minded thinking as a case study. *Cognition*,

187(February), 156–166. <https://doi.org/10.1016/j.cognition.2019.03.006>

Stanovich, K. E., & Toplak, M. E. (2023). Actively Open-Minded Thinking and Its Measurement. *Journal of Intelligence*, 11(2), 1–22. <https://doi.org/10.3390/jintelligence11020027>

Suparman, A., Samsudin, A., Kaniawati, I., & Lantik, V. (2023). Analysis of students' critical thinking skill on optical instruments through worksheet-based inductive learning model. *AIP Conference Proceedings*, 2734(1). <https://doi.org/10.1063/5.0155392>

Svedholm-Häkkinen, A. M. (2009). Actively open-minded thinking : development of a shortened scale and disentangling attitudes towards knowledge and people. *Thinking & Reasoning*, 24(1), 21–40.

<https://doi.org/https://doi.org/10.1080/13546783.2017.1378723>

<http://hdl.handle.net/10138/311318>

Svedholm-Häkkinen, A. M., & Lindeman, M. (2018). Actively open-minded thinking: development of a shortened scale and disentangling attitudes towards knowledge and people. *Thinking & Reasoning*, 24(1), 21–40. <https://doi.org/10.1080/13546783.2017.1378723>

Tanyeli Zeki, N., & Kuter, S. (2018). Impact of collaborative and reflective writing activities on students' autonomy in writing. *Quality and Quantity*, 52(s1), 343–360. <https://doi.org/10.1007/s11135-017-0615-3>

Tkavashvili, E. (2021). the Impact of Teacher Professional Diary on Their Reflective, Communicative and Professional Skills. *Problems of Education in the 21st Century*, 79(2), 273–295. <https://doi.org/10.33225/pec/21.79.273>

Toplak, M. E., West, R. F., & Stanovich, K. E. (2014). Rational thinking and cognitive sophistication: Development, cognitive abilities, and thinking dispositions. *Developmental Psychology*, 50(4), 1037–1048. <https://doi.org/10.1037/a0034910>

van Velzen, J. H. (2015). Are students intentionally using self-reflection to improve how they learn? Conceptualising self-induced self-reflective thinking. *Reflective Practice*, 16(4), 522–533. <https://doi.org/10.1080/14623943.2015.1064378>

Wallace, J. P., Blinkhorn, A. S., Blinkhorn, F. A., & Wallace, J. P. (2013). *Reflective folios for dental hygiene students : what do they tell us about a residential aged care student placement experience ?* 17, 236–240. <https://doi.org/10.1111/eje.12044>

Wong, F. K., Kember, D., Chung, L. Y. F., & CertEd, L. Y. (1995). Assessing the level of student reflection from reflective journals. *Journal of Advanced Nursing*, 22(1), 48–57. <https://doi.org/10.1046/j.1365-2648.1995.22010048.x>

Yan, S., Sun, D., Zhou, Q., Yang, Y., & Tian, P. (2023). Exploring the impact of virtual laboratory with KWL reflective thinking approach on students' science learning in higher education. *Journal of Computing in Higher Education*. <https://doi.org/10.1007/s12528-023-09385-y>

Yuen Lie Lim, L.-A. (2011). A comparison of students' reflective thinking across different years in a problem-based learning environment. *Instructional Science*, 39(2), 171–188. <https://doi.org/10.1007/s11251-009-9123-8>

Zach, S., & Ophir, M. (2020). Using simulation to develop divergent and reflective thinking in teacher education. *Sustainability (Switzerland)*, 12(7). <https://doi.org/10.3390/su12072879>

Zhai, N., Huang, Y., Ma, X., & Chen, J. (2023). Can reflective interventions improve students' academic achievement? A meta-analysis. *Thinking Skills and Creativity*, 49(July), 101373. <https://doi.org/10.1016/j.tsc.2023.101373>

Pipih Nurhayati
Science Education Program, Universitas Pendidikan Indonesia
Bandung, Jawa Barat, Indonesia
pipih.nurhayati@upi.edu

Dear Editorial Team,

Finding the right journal for our work has been a challenging journey, but we are grateful for the guidance that led us to transfer our article to Social Sciences & Humanities Open. We believe this journal is the appropriate platform to showcase our research.

My focus is on filling the research gap regarding the mediators of reflective thinking skills, which ultimately converge on open-mindedness. My manuscript is titled "Opening Minds, Fostering Reflection: A Pathway Analysis of Open-Mindedness and Reflective Thinking Skills in Science Education". Beginning with Dewey's philosophy and further reading on the necessity of humility in the learning process, especially in science education, I was motivated to empirically investigate the relationship between open-mindedness and reflective thinking skills in science education.

To answer this question, I employed correlation analysis, pathway analysis, and regression between each indicator of Actively Open-minded Thinking and Reflective Thinking Skills. This study provides empirical evidence that open-mindedness, as measured by actively open-minded thinking, influences students' understanding, reflection, and critical reflection in science education.

With the ongoing development of research on predictors of thinking skills, this manuscript will contribute to the literature for science education developers, thinking skills practitioners, and educational psychology.

This work is original and has not been published elsewhere.

Thank you for your attention and consideration. I look forward to hearing from you.

Yours sincerely,

Pipih Nurhayati
Science Education Program, Universitas Pendidikan Indonesia
Bandung, Jawa Barat, Indonesia

Opening Minds, Fostering Reflection: A Pathway Analysis of Open-Mindedness and Reflective Thinking Skills in Science Education

Pipih Nurhayati^{1*}, Ari Widodo², Widi Purwianingsih³, Achmad Samsudin⁴

¹⁻² Program of Science Education, Universitas Pendidikan Indonesia, Bandung, Indonesia

³ Program of Biology Education, Universitas Pendidikan Indonesia, Bandung, Indonesia

⁴ Program of Physics Education, Universitas Pendidikan Indonesia, Bandung, Indonesia
pipih.nurhayati@upi.edu

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Ethics Statement

This study was conducted in strict accordance with the ethical standards laid out in the Declaration of Helsinki. All procedures involving human participants were approved and informed consent was obtained from all participants before they were included in the study. Participants were provided with detailed information regarding the purpose of the research, their role, and the confidentiality of their data. They were also informed of their right to withdraw from the study at any time without any consequences.